School Worksheet

Class 12 Mathematics, Chapter 8: Application of Integrals

Maximum marks: **53** **20** questions

Name _____ Date _____ Score _____ / 53

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

The area of the region bounded by $y = x^2$, the x-axis and the lines $x = 0$ and $x = 3$ is:

(A) 3 sq units **(B)** 9 sq units

(C) 27 sq units **(D)** 6 sq units

2 MCQ — 1 mark

The area enclosed by the circle $x^2 + y^2 = 4$ is:

(A) 2π sq units **(B)** 4π sq units

(C) 16π sq units **(D)** π sq units

3 MCQ — 1 mark

The area enclosed by the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is:

(A) 6π sq units **(B)** 36π sq units

(C) 13π sq units **(D)** 12π sq units

4 MCQ — 1 mark

The area of the region bounded by $y = \cos x$, the x-axis, $x = 0$ and $x = \frac{\pi}{2}$ is:

(A) 2 sq units **(B)** 1 sq unit **(C)** $\frac{\pi}{2}$ sq units **(D)** 0

5 MCQ — 1 mark

The area of the region bounded by the curve $y = x$ and the x-axis, between $x = 0$ and $x = 2$, is:

(A) 2 sq units **(B)** $\frac{1}{2}$ sq unit **(C)** 1 sq unit **(D)** 4 sq units

Q No.	Question	Marks
6	Assertion-Reason **Assertion (A):** The area bounded by $y = \sin x$ and the x-axis between $x = 0$ and $x = 2\pi$ is 4 sq units. **Reason (R):** $\int_0^{2\pi} \sin x \, dx = 0$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer Find the area of the region bounded by $y^2 = x$, the x-axis and the lines $x = 1$ and $x = 4$, lying in the first quadrant.	2 marks
Level 2		**18 marks**
8	Short answer Find the area of the region bounded by $y^2 = 9x$, $x = 2$, $x = 4$ and the x-axis in the first quadrant.	3 marks
9	Short answer Find the area of the region bounded by $x^2 = 4y$, $y = 2$, $y = 4$ and the y-axis in the first quadrant.	3 marks
10	Short answer Using integration, find the area of the region bounded by the line $y = 3x + 2$, the x-axis and the ordinates $x = -1$ and $x = 1$.	3 marks
11	Short answer Find the area of the region in the first quadrant bounded by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the coordinate axes.	2 marks
12	Short answer Find the area of the region bounded by the circle $x^2 + y^2 = 16$, the x-axis and the lines $x = 0$ and $x = 2$, lying in the first quadrant.	3 marks
13	Short answer Find the area bounded by $y = \sin x$ and the x-axis between $x = 0$ and $x = \frac{3\pi}{2}$.	2 marks
14	Short answer Find the area of the region bounded by $y = x^3$, the x-axis and the lines $x = -2$ and $x = 1$.	2 marks

Q No.	Question	Marks
Level 3		**27 marks**
15	Long answer Find the area of the region in the first quadrant enclosed by the x-axis, the line $y = x$ and the circle $x^2 + y^2 = 32$.	5 marks
16	Long answer Using integration, find the area of the triangle whose vertices are $(1, 0)$, $(2, 2)$ and $(3, 1)$.	5 marks
17	Long answer Find the area of the region bounded by the curve $y = 4x^2$ and the line $y = 8x + 12$.	5 marks
18	Long answer Using integration, find the area of the region bounded by the curve $4x^2 + y^2 = 36$.	5 marks
19	Short answer Find the area of the region bounded by the parabola $y^2 = 4ax$ and its latus rectum.	3 marks
20	Case study **Case Study:** A running track in a sports complex is in the shape of the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$ (units in metres). The region inside the track is to be grassed. **(i)** Find the total area enclosed by the track. — 1 mark **(ii)** Find the area of the region in the first quadrant enclosed by the track and the axes. — 1 mark **(iii)** Find the area of the region inside the track between $x = 0$ and $x = 3$ in the first quadrant. — 2 marks	4 marks

Answer key

Chapter 8: Application of Integrals, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) 9 sq units
2	(B) 4π sq units
3	(A) 6π sq units
4	(B) 1 sq unit
5	(A) 2 sq units
6	(B) Both A and R are true, but R is not the correct explanation of A
7	$\frac{14}{3}$ sq units

Level 2

8	$16 - 4\sqrt{2}$ sq units
9	$\frac{32 - 8\sqrt{2}}{3}$ sq units
10	$\frac{13}{3}$ sq units
11	3π sq units
12	$2\sqrt{3} + \frac{4\pi}{3}$ sq units
13	3 sq units
14	$\frac{17}{4}$ sq units

Level 3

15	4π sq units
16	$\frac{3}{2}$ sq units
17	$\frac{128}{3}$ sq units
18	18π sq units
19	$\frac{8a^2}{3}$ sq units
20	(i) $24\pi\ \mathrm{m}^2$ (ii) $6\pi\ \mathrm{m}^2$ (iii) $\left(3\sqrt{3} + 2\pi\right)\ \mathrm{m}^2$